Experiment – 05

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Question 1: User Confirmation Rate (Medium)

A website maintains records of users who have registered on the platform and the status of confirmation requests sent to them. A confirmation request can either be successfully completed (confirmed) or may expire without completion (timeout). For each registered user, calculate the confirmation rate, defined as the number of successful confirmations divided by the total number of confirmation requests sent. If a user has never received any confirmation request, their confirmation rate should be considered 0. Display the user ID along with the confirmation rate rounded to 2 decimal places. The result can be returned in any order.

Signups Table:

user_id	time_stamp
3	2020-03-21 10:16:13
7	2020-01-04 13:57:59
2	2020-07-29 23:09:44
6	2020-12-09 10:39:37

Confirmations Table:

user_id	time_stamp	action
3	2021-01-06 03:30:46	timeout
3	2021-07-14 14:00:00	timeout
7	2021-06-12 11:57:29	confirmed
7	2021-06-13 12:58:28	confirmed
7	2021-06-14 13:59:27	confirmed
2	2021-01-22 00:00:00	confirmed
2	2021-02-28 23:59:59	timeout

Example Output:

user_id	confirmation_rate
2	0.50
3	0.00
6	0.00
7	1.00

Solution:

```
SELECT
  S.USER_ID,
  ROUND(
    COALESCE(
      SUM(CASE WHEN C.ACTION = 'confirmed' THEN 1 ELSE 0 END) * 1.0
      / NULLIF(COUNT(C.ACTION), 0),
      0
    ), 2
  ) AS CONFIRMATION_RATE
FROM SIGNUPS AS S
LEFT JOIN CONFIRMATIONS AS C
  ON S.USER_ID = C.USER_ID
GROUP BY S.USER_ID;
```

Question 2: Trips and Cancellation Rate (Hard)

A ride-booking company maintains information about trip requests and the users involved in those trips. Some users may be restricted (banned) from using the platform, while others are active participants. Each trip can either be completed successfully or cancelled by one of the participants. For each day between 2013-10-01 and 2013-10-03, determine the proportion of cancelled trips among all valid trips. A trip is valid only when both the customer and the driver associated with that trip are active (not banned) users. Display the date and the cancellation rate rounded to two decimal places.

Users Table:

users_id	banned	role
1	No	client
2	Yes	client
3	No	client
4	No	client
10	No	driver
11	No	driver
12	No	driver
13	No	driver

Trips Table:

id	client_id	driver_id	city_id	status	request_at
1	1	10	1	completed	2013-10-01
2	2	11	1	cancelled_by_driver	2013-10-01
3	3	12	6	completed	2013-10-01
4	4	13	6	cancelled_by_client	2013-10-01
5	1	10	1	completed	2013-10-02
6	2	11	6	completed	2013-10-02
7	3	12	6	completed	2013-10-02
8	2	12	12	completed	2013-10-03

9	3	10	12	completed	2013-10-03
10	4	13	12	cancelled_by_driver	2013-10-03

Example Output:

Day	Cancellation Rate
2013-10-01	0.33
2013-10-02	0.00
2013-10-03	0.50

Solution:

```
SELECT
  T.REQUEST_AT AS DAY,
  ROUND(
    SUM(CASE WHEN T.STATUS IN ('cancelled_by_driver','cancelled_by_client')
             THEN 1 ELSE 0 END) * 1.0 / COUNT(*), 2
  ) AS CANCELLATION_RATE
FROM TRIPS AS T
JOIN USERS AS UC
  ON T.CLIENT_ID = UC.USERS_ID AND UC.BANNED = 'No'
JOIN USERS AS UD
  ON T.DRIVER_ID = UD.USERS_ID AND UD.BANNED = 'No'
WHERE T.REQUEST_AT BETWEEN '2013-10-01' AND '2013-10-03'
GROUP BY T.REQUEST_AT
ORDER BY T.REQUEST_AT;
```